

## Diego González Sánchez

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### Employment

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|-------------------|-------------------------------------------------------------------------|
| 11/2020 – Present | <b>Research fellow</b> , Alfréd Rényi Institute of Mathematics, Hungary |
| 02/2024 – 05/2024 | <b>Professor</b> , Budapest Semesters in Mathematics, Hungary           |

### Education

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| 10/2016 – 10/2020 | <b>Ph.D.</b> , Autonomous University of Madrid, Spain<br>Thesis: <i>Topics in additive combinatorics and higher order Fourier analysis</i><br>Supervisor: Prof. Pablo Candela. Defended on 25/09/2020.<br>Final score: Outstanding ( <i>sobresaliente</i> ) cum laude.      |
| 09/2015 – 07/2016 | <b>M.A.St.</b> , in Pure Mathematics (Part III), Cambridge University, UK<br>Final score: 73/100, Merit                                                                                                                                                                     |
| 09/2009 – 06/2015 | <b>B.Sc.</b> in Mathematics, Autonomous University of Madrid, Spain.<br>Final score: 9.41/10 outstanding ( <i>sobresaliente</i> )<br><b>B.Sc.</b> in Computer Science, Autonomous University of Madrid, Spain.<br>Final score: 9.41/10 outstanding ( <i>sobresaliente</i> ) |

### Visiting periods (at least one month)

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| 05/2019 – 08/2019 | Visiting researcher, Alfred Renyi Institute of Mathematics, Hungary |
| 09/2013 – 07/2014 | Erasmus student, Technical University of Denmark, Denmark           |

### Papers published or accepted for publication

- [1] *On  $\mathbb{F}_2^\omega$ -affine-exchangeable probability measures*, (with P. Candela and B. Szegedy). *Studia Mathematica*, (2024), 79 (1) . Open access: [arXiv:2203.08915](https://arxiv.org/abs/2203.08915).
- [2] *On measure-preserving  $\mathbb{F}_p^\omega$  systems of order  $k$* , (with P. Candela and B. Szegedy). *Journal d'Analyse Mathématique (JAMA)*, (2025). Open access: [arXiv:2311.13899](https://arxiv.org/abs/2311.13899).
- [3] *On higher-order Fourier analysis in characteristic  $p$* , (with P. Candela and B. Szegedy). *Ergodic Theory and Dynamical Systems*, 1-70, (2022). Open access: [arXiv:2109.15281](https://arxiv.org/abs/2109.15281).
- [4] *A refinement of Cauchy-Schwarz complexity*, (with P. Candela and B. Szegedy). *European Journal of Combinatorics* 106, 103592 (2022). Open access: [arXiv:2109.05965](https://arxiv.org/abs/2109.05965).
- [5] *New estimates for exponential sums over multiplicative subgroups and intervals in prime fields*, (with D. Di Benedetto, M. Z. Garaev, V. C. Garcia, I. E. Shparlinski, and C. A. Trujillo). *Journal of Number Theory* 215, 261-274 (2020). Open access: [arXiv:2003.06165](https://arxiv.org/abs/2003.06165).
- [6] *On sets with small sumset and  $m$ -sum-free sets in  $\mathbb{Z}/p\mathbb{Z}$* , (with P. Candela and D. J. Grynkiewicz. *Bulletin de la Société Mathématique de France* 149 (1), 155-177 (2020). Open access: [arXiv:1909.07967](https://arxiv.org/abs/1909.07967).
- [7] *A note on the Bilinear Bogolyubov Theorem: Transverse and bilinear sets*. With P.-Y. Bienvenu and A. D. Martínez. *Proceedings of the American Mathematical Society* 148 (1), 23-31 (2020). Open access: [arXiv:1811.09853](https://arxiv.org/abs/1811.09853).

- [8] | *On nilspace systems and their morphisms.*, (with P. Candela and B. Szegedy). Ergodic Theory and Dynamical Systems, Volume 40, Issue 11, (2020) , pp. 3015 – 3029 . Open access: arXiv:1807.11510.
- [9] | *A Plünnecke-Ruzsa inequality in compact abelian groups.*, (with P. Candela and A. de Roton). Revista Matemática Iberoamericana 35 (7), 2169-2186 (2019). Open access: arXiv:1712.07615.

## Conference papers

- [10] | *Higher-order Fourier analysis via spectral algorithms*, (with P. Candela and B. Szegedy). Proceedings of the 13th European Conference on Combinatorics, Graph Theory and Applications EUROCOMB'25 Budapest, August 25–29, (2025). Open access: arXiv:2501.12287.
- [11] | *A short proof of an inverse theorem in bounded torsion groups*, (with P. Candela and B. Szegedy). Discrete Mathematics Days, Alcalá de Henares, July 3-5, (2024). Open access: arXiv:2311.13899.
- [12] | *Optimal transport with  $f$ -divergence regularization and generalized Sinkhorn algorithm*, (with D. Terjék). International Conference on Artificial Intelligence and Statistics, 5135-5165 (2022). Open access: arXiv:2105.14337.
- [13] | *A Refinement of Cauchy-Schwarz Complexity, with Applications*, (with P. Candela and B. Szegedy). Extended Abstracts EuroComb 2021, 293-298 (2021). Open access: arXiv:2109.05965.
- [14] | *A step towards the  $3k - 4$  conjecture in  $\mathbb{Z}/p\mathbb{Z}$  and an application to  $m$ -sum-free sets*, (with P. Candela and D. J. Gryniewicz). Acta Mathematica Universitatis Comenianae, Volume 88, (2019), Issue 3, pp. 521-525 (2019). Open access: arXiv:1909.07967.

## Preprints

- [15] | *MLPs at the EOC: Concentration of the NTK*, (with D. Terjék). Open access: arXiv:2501.14724.
- [16] | *MLPs at the EOC: Spectrum of the NTK*, (with D. Terjék). Open access: arXiv:2501.13225.
- [17] | *Spectral algorithms in higher-order Fourier analysis*, (with P. Candela and B. Szegedy). Open access: arXiv:2501.12287.
- [18] | *On the inverse theorem for Gowers norms in abelian groups of bounded torsion*, (with P. Candela and B. Szegedy). Open access: arXiv:2311.13899.
- [19] | *Free nilspaces, double-coset nilspaces, and Gowers norms*, (with P. Candela and B. Szegedy). Open access: arXiv:2305.11233.
- [20] | *A framework for overparameterized learning*, (with D. Terjék). Open access: arXiv:2205.13507.

## Contributed talks at conferences and workshops

- 08/2025 | *Higher-order Fourier analysis via spectral algorithms* at EuroComb 2025, Budapest, Hungary.
- 07/2025 | *Inverse Gowers Theory via Ergodic Structure Theory* (together with Giorgio Baglioni and Nuno Carneiro) at 28th International Internet Seminar (ISEM28) “Ergodic structure theory and applications”, Wuppertal, Germany.

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| 07/2024 | <i>A short proof of an inverse theorem in bounded torsion groups</i> at Discrete Mathematics Days 2024, Alcalá de Henares, Spain.                                                               |
| 11/2023 | <i>On measure-preserving <math>\mathbb{F}_p^\omega</math> systems of order <math>k</math></i> at the 11th workshop Operator Theoretic Aspects of Ergodic Theory, Wuppertal, Germany.            |
| 06/2023 | <i>Free nilspaces, double-coset nilspaces, and Gowers norms</i> at Nilpotent structures in topological dynamics, ergodic theory and combinatorics, Będlewo Conference Center, Poland.           |
| 05/2022 | <i>On affine-exchangeable measures and nilspaces</i> at the workshop Graphs, Groups, Stochastic Processes, Alfréd Rényi Institute of Mathematics, Budapest, Hungary.                            |
| 09/2021 | <i>A Refinement of Cauchy-Schwarz Complexity, with Applications</i> , at EuroComb 2021, online (Barcelona, Spain).                                                                              |
| 08/2019 | <i>The <math>3k - 4</math> theorem in <math>\mathbb{Z}/p\mathbb{Z}</math> and an application to <math>m</math>-sum-free sets</i> at EuroComb 2019, Univerzita Komenského, Bratislava, Slovakia. |
| 12/2017 | <i>La desigualdad de Plünnecke-Ruzsa en grupos abelianos compactos</i> at the LXVI Reunión de Comunicaciones Científicas of RSME and UMA, Buenos Aires, Argentina.                              |

## Invited talks at conferences and workshops

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| 09/2025 | <i>A spectral approach to higher-order Fourier analysis, Part I</i> at Approximate Algebraic Structures and Arithmetic Combinatorics, Mathematical Institute of the Serbian Academy of Sciences and Arts, Belgrade, Serbia. |
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## Invited talks at seminars

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| 02/2025 | <i>On higher-order Fourier analysis</i> at Department of Mathematics and Department of Quantitative Methods seminar, CUNEF universidad, Spain.                                                                                          |
| 11/2024 | <i>An algorithm for higher-order Fourier analysis</i> at Séminaire de Théorie des Nombres de Nancy-Metz, Institut Élie Cartan de Lorraine, France.                                                                                      |
| 06/2024 | <i>On measure-preserving <math>\mathbb{F}_p^\omega</math>-systems of order <math>k</math></i> at DAGGER seminar, University of Warwick, UK.                                                                                             |
| 09/2022 | <i>The Polyak-Łojasiewicz condition and overparameterized learning</i> (with D. Terjék) at Euler Seminar, Université catholique de Louvain, Louvain, Belgium.                                                                           |
| 03/2019 | <i>Conjuntos grandes que se parecen a progresiones en grupos cíclicos de orden primo</i> (Large sets that resemble arithmetic progressions in cyclic groups of prime order), at Junior Seminar, Autonomous University of Madrid, Spain. |
| 06/2018 | <i>A Plünnecke-Ruzsa inequality in compact abelian groups</i> at UAM-ICMAT number theory seminar, ICMAT, Madrid, Spain.                                                                                                                 |
| 03/2017 | <i>El método polinomial en combinatoria aditiva</i> (the polynomial method in additive combinatorics) at the Junior Seminar, Autonomous University of Madrid, Spain.                                                                    |

## Other events attended

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| 07/2023 | 200 Years of Trinity Combinatorics, Cambridge, UK.                                    |
| 06/2022 | Graphs, Groups, Stochastic processes summer school, Budapest, Hungary.                |
| 08/2019 | Number theory in the Americas. Casa Matemática Oaxaca, Oaxaca, Mexico.                |
| 09/2018 | Arithmetic Ramsey Theory, University of Manchester, Manchester, UK.                   |
| 09/2017 | The music of Numbers. Conference in honor of Javier Cilleruelo. ICMAT, Madrid, Spain. |

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| 05/2017 | Graduate course on Interactions of harmonic analysis, combinatorics and number theory. BGSMATH, Barcelona, Spain. |
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## Events organized

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| 06/2025 | New Trends in Arithmetic Combinatorics and related Fields, a BIRS-IMAG workshop. Coorganized with: P. Candela (lead organizer), A. de Roton, H. Helfgott, A. Sedunova, and O. Serra. Check our website: <a href="https://birs-imag-ntac.github.io/">https://birs-imag-ntac.github.io/</a> . |
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## Projects in which I have participated and role

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| 03/2022 – 12/2025 | <i>Artificial Intelligence National Laboratory (MILAB, RRF-2.3.1-21-2022-00004)</i> , NKFIH, Hungary. It partially funds the Research Fellow position at Alfréd Rényi institute of Mathematics. I work in the mathematical foundations of AI. |
| 09/2021 – 08/2024 | <i>Harmonic analysis, combinatorics and arithmetic</i> (PID2020-113350GB-I00) funded by MICINN, Spain. I am part of the working team.                                                                                                         |
| 11/2020 – 11/2022 | <i>Momentum (Lendület) 30003</i> , NKFIH, Hungary. It partially funds the Research Fellow position at Alfréd Rényi institute of Mathematics. I work in HOF and in the mathematical foundations of AI.                                         |
| 11/2020 – 03/2025 | <i>“Élvonal” KKP 133921 grant</i> , NKFIH, Hungary. It partially funds the Research Fellow position at Alfréd Rényi institute of Mathematics. I work in HOF and deep learning in this project.                                                |
| 01/2018 – 12/2020 | <i>Arithmetic and harmonic analysis</i> , (MTM2017-83496-P) MINECO, Spain. I am part of the working team.                                                                                                                                     |

## Research Fellowships/Prizes/Grants

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| 15/07/2021 | Extraordinary award to the best theses presented at the Autonomous University of Madrid in the period 01/10/2019 to 30/09/2020. Awarded by the Autonomous University of Madrid.                        |
| 10/02/2017 | La Caixa Severo Ochoa scholarship awarded by Fundación la Caixa (private company). Ph.D.’s funding. It included a monthly stipend, an amount for travels and other expenses, and all labour/tax costs. |
| 27/06/2015 | X Mutua Madrileña Postgraduate Scholarship awarded by Fundación Mutua Madrileña (private company). M.A.St.’s funding.                                                                                  |
| 18/06/2014 | JAE-intro scholarship for introduction to research awarded by CSIC, Ministry of Economy and Competitiveness, Spain.                                                                                    |
| 03/12/2012 | Madrid Excellence Scholarship awarded by the Autonomous Community of Madrid, Spain.                                                                                                                    |
| 14/09/2011 | Madrid Excellence Scholarship awarded by the Autonomous Community of Madrid, Spain.                                                                                                                    |
| 26/01/2010 | Madrid Excellence Scholarship awarded by the Autonomous Community of Madrid, Spain.                                                                                                                    |

## Accreditations

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| 12/01/2022 | <i>Profesor ayudante doctor</i> (assistant professor) granted by the Ministry of Universities of Spain. |
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## Graduate teaching

08/2021 | Instructor at AGRA IV (Arithmetic, Groups and Analysis), online (Trieste, Italy).

## Undergraduate teaching

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| 02/2024 – 05/2024 | Professor (jointly with P. Zsámboi) of <i>Introduction to Deep Learning</i> at the Budapest Semesters in Mathematics (BSM), Spring Semester 2024 (28 hours taught). Note: This was the first time this course was taught at BSM. Thus, Zsámboi and I had to prepare it from scratch. |
| 09/2019 – 11/2019 | Assistant professor of <i>Mathematics I</i> for the B.Sc. in Chemical engineering at the Autonomous University of Madrid (6 hours taught).                                                                                                                                           |
| 09/2018 – 01/2019 | Assistant professor of <i>Mathematics</i> for the B.Sc. in Environmental Sciences at the Autonomous University of Madrid (15 hours taught).                                                                                                                                          |
| 09/2017 – 01/2018 | Assistant professor of <i>Algebra I</i> for the B.Sc. in Physics at the Autonomous University of Madrid (15 hours taught).                                                                                                                                                           |

## Service

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| Referee for  | TEMat, <i>Analysis and PDE</i> , <i>AISTATS 2022</i> , <i>AISTATS 2024</i> and <i>The Ramanujan Journal</i> . |
| Reviewer for | <i>Zentralblatt MATH (zbMath)</i> .                                                                           |

## Other outreach activities

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| 22/04/2024 | <i>Recent trends III: Subsets of the integers without three term arithmetic progressions</i> , an outreach article for the Discrete and Algorithmic Mathematics (DAM) network. Available at: <a href="https://dam-network.github.io/">https://dam-network.github.io/</a> . |
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## Languages

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| Human:       | Spanish (mother tongue)<br>English (fluent, used in professional contexts)<br>French (basic). |
| Programming: | Python (including Pytorch), C, Java, SQL, Matlab, and R.                                      |